Matt | Varghese Mathew

Software Systems Architect

Software Systems Architect with 15+ years of hands-on experience engineering secure, multi-vendor distributed platforms and real-time API orchestration. Former UW-Madison Condor Research Assistant with a strong background in High Throughput Computing and deterministic runtime environments. Proven track record of translating complex requirement-specifications into production-ready systems for hundreds of enterprise clients. Strong interest in GNU/Linux, Open-Source, and contributing back to Condor!

Ask Me || DM me on LinkedIn
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CONDOR WORK EXPERIENCE

Graduate Research Assistant, Condor HTC

University of Wisconsin - Madison, WI *SEP 2008 - MAY 2010*

Under the advisement of Prof. Miron Livny, engineered deterministic runtime environments by virtualizing Condor jobs within VMware, developing early (pre-docker) patterns for Infrastructure-as-a-Service (IaaS). Investigated two distinct virtual machine job-delivery patterns to achieve reliable execution across heterogeneous environments.

- **Self-Contained Execution:** Packaged jobs directly within VMs to run upon startup.
- **Autonomous Pool Integration:** Dispatched VMs that dynamically joined the Condor pool at runtime to claim their own designated workloads.

PROJECTS SPOTLIGHT

Personal HTCondor Cluster *2016*

Designed and implemented a 10-node Raspberry Pi 3B cluster running HTCondor. Managed cluster and network architecture to simulate high-throughput computing workloads in a hardware-constrained ARM environment.

MIPS Microprocessor Pipeline Simulator *2005, 2025*

github.com/mattvarghese/coconut

5-Stage pipeline simulation of a MIPS microprocessor, with cache simulation, assembler, and terminal app developed with POSIX APIs, C++, Flex, and Bison in 2005. In 2025, developed a Small-C compiler above the assembler with AI assistance. Demonstrates expertise in computer architecture, parallelism, and performance.

RFC-compliant JSON Serializer & Deserializer for ANSI-M *2020*

Created RFC 8259 compliant generic JSON Serializer and Deserializer for the ANSI-M language. Library enables 30+ database AI integrations and analytics across Epic, spotlighting architectural vision in creating powerful generic reusable tooling.

EDUCATION

Master of Science, Computer Science

University of Wisconsin - Madison *2008-2010*

GPA: 3.94 / 4.0 | Focus: Systems, Networking, Distributed Computing.

Bachelor of Technology, Computer Science and Engineering

National Institute of Technology - Calicut, India *2002-2006*

CGPA: 9.60 / 10.0 | Department Topper and Gold Medalist.

SKILLS

Architecture & Infrastructure

- SDLC
- Distributed Systems
- Microservices
- Azure
- CI/CD
- Docker
- Virtualization
- Hadoop
- Maintainability
- Performance

Engineering Stack

- C/C++
- Python
- Git & SVN
- Web API
- TypeScript
- React
- Node
- .NET/C#
- ANSI-M
- SQL
- Linux
- Windows

Specialized Expertise

- LLM API Integration
- Agentic Development
- Regulatory Infrastructure
- Network Architecture & Routing

COLLABORATION

Standards-body Hackathons

Developed prototypes of industry specifications in collaboration with participants from other vendors to improve interoperability.

Authoring Industry Standards

Contributed to two emerging industry specifications for interoperability towards improving workflow integration & security.

Steering Industry Standards

Represented Epic on public calls steering development of three standards to reduce administrative overhead for high-latency B2B transaction approvals.

WORK EXPERIENCE

Software Systems Architect, Interoperability

Epic - Verona, WI

MAR 2021 - PRESENT

Architect for a large-scale, standards-based, regulatory integration platform, defining automated real-time decision workflows for enterprise-scale organizations. Engineered systems adopted by hundreds of major enterprise clients, reducing high-latency business process turnarounds from several days to near real-time.

- **Led architecture and delivery of a regulatory interoperability platform** for real-time data exchange, collaborating with standards bodies and external vendors to transform emerging specifications into scalable, production-ready infrastructure.
- **Identified gaps and established a unified "source of truth"** replacing fragmented logic across 12+ product modules, reducing lead time for new feature development by ~80%, and improving long-term maintainability.
- **Recognized the need and architected standardized** data models and execution frameworks replacing inconsistent one-off implementations for maintainabilityy.
- **Modernized third-party decision-engine integrations** using OAuth2 SSO patterns, enabling real-time workflow execution and reducing integration defects by ~60%.
- **Mitigated tooling deficiencies by building web applications** to emulate third-party systems, validating workflows end-to-end, and accelerating partner onboarding.

Senior Software Systems Developer, Decision Support

Epic - Verona, WI

AUG 2010 - MAR 2021

Senior developer responsible for distributed decision-support infrastructure, external web application integrations, and mobile push notifications services supporting large-scale enterprise applications. Directed adoption of modern interoperability standards and pioneered Epic's first high-availability cloud service on Azure.

- **Created Epic's high-availability mobile push notifications platform,** routing millions of weekly notifications via APNS and GCM. Pioneered deployment as one of Epic's first Azure-native production services, reducing operational costs by over 90%.
- **Identified fragmentation and led enterprise-wide centralization** of 3rd-party web application integrations to establish standards for security and maintainability. Developed a suite of validation apps for CSP, CORS, and Web Messaging APIs.
- **Designed & Developed Epic's decision support web services framework,** enabling reusable standards-based workflow integrations across multiple enterprise applications using RESTful APIs and event-driven execution patterns.
- **Identified usability gaps and redesigned** decision support configuration to align with mental models, improving usability and enabling efficacy feedback loops for users.

Software Engineer, Enterprise Routing

Cisco Systems - Bangalore, India

JUN 2006 - JUL 2008

Platform engineering and configuration tooling for Cisco's enterprise class routers.

COLLABORATION continued...

Epic Vendor Conferences

Presented for several years on Epic's web service integration capabilities to educate and onboard third-party vendors.

Epic Mentoring

Mentored engineers and interns across multiple teams, driving adoption of architectural best practices and development of high-performing teams.

Epic Recruitment

Attracted top talent to Epic by representing Epic at university career fairs and conducting candidate interviews.

EXPLORATORY

LLM-API Workflow Assistant

2025

Single-sign-on AI assistant web application.

Interfaces LLM APIs with enterprise data sources to support real-time chat-based interaction for in-workflow assistance.

Web Based SSH Client

2025

Non-traditional solution for real world cost reduction avoiding remote-app licenses for moderated shell access to servers.

Demonstrated proof of concept fully browser-based SSH client for unix servers.

github.com/mattvarghese/WebSSHClient

HTTP Auto-responder

2024

Development tooling for Web APIs, keeping traffic within the enterprise or local network.

Configurable endpoint allows simultaneous multi-user testing of RESTful APIs.

github.com/mattvarghese/HttpResponder

OTHER

- USHPA Novice (H2) rated hang-glider pilot
- Solo endorsed VFR student pilot
- ASA 101,103/104,105 rated sailor
- Learned Elementary Russian as an adult
- Avid reader, ice-skating, volleyball, XC-skiing, bicycling